

Daily Geometry Drill #7

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Loci of points · Apollonius circle $\frac{PA}{PB} = k$ · geometry-logic conditionals · flaw-spotting in synthetic proofs.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Name the locus straight off; only the standard families occupy these questions.

- Write down the locus of points whose distance from the origin is 5.
- Write down the locus of points equidistant from the two points $(0,0)$ and $(4,0)$.
- The locus of P with $PA = 2PB$, where $A = (0,0)$ and $B = (6,0)$, is a circle. Write down the x -coordinate of its (relevant) endpoint on the x -axis where $y = 0$ between A and B .
- Write down the locus of points that are always 3 units from the fixed line $y = 4$.
- The locus of points whose distance to the fixed point $(0,1)$ equals their distance to the fixed line $y = -1$ is a parabola. Write down a point on it other than the origin with integer coordinates.
- The locus of points whose distance to the point $(1,2)$ is exactly 3 is a circle. Write down its equation.
- Write down the locus of the midpoint M of every radius OQ of the circle $x^2 + y^2 = 4$ as Q runs around the circle (with O the origin).
- The equation $x^2 + y^2 = 2x$ describes a circle. Write down its radius.
- Write down whether the statement “ $x^2 + y^2 = 4$ implies $x = \pm 2$ or $y = \pm 2$ ” is true or false.
- Write down the locus of points P with $PA = PB$, where $A = (1,3)$ and $B = (5,3)$. (Give its equation in x .)

Section B Manipulation Drills

(≤50 s each)

Square both sides to force Apollonius loci into circle form, or reason about necessity/sufficiency for the logic items.

- The circle of Apollonius for $PA = 2PB$ with $A = (0,0)$, $B = (6,0)$ has radius?
A) 4

- B) 6
C) 8
D) 12
2. Locus of points equidistant from $(0, 0)$ and $(0, 4)$ is?
A) $x = 2$
B) $y = 2$
C) $y = 4$
D) $x = 4$
3. The points satisfying $x^2 + y^2 \leq 4$ form a filled disk. Its area is?
A) 2π
B) 4π
C) 8π
D) 16
4. The distance from $P(x, y)$ to the origin is $\sqrt{x^2 + y^2}$. The locus $\sqrt{x^2 + y^2} = 1$ is a circle. Write down its circumference.
5. A point is known to lie strictly inside the unit disk $x^2 + y^2 < 1$. Which of the following statements is *necessarily* true for such a point?
A) $x^2 + y^2 < \frac{1}{2}$
B) $x + y < 1$
C) $x^2 + y^2 < 2$
D) $x^2 + y^2 > 0$
6. The Apollonius locus $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$: what is the x -coordinate of its right endpoint on the x -axis?
A) 3
B) 4
C) 6
D) 9
7. To prove a point P lies on the unit circle $x^2 + y^2 = 1$, which single condition is *sufficient* on its own?
A) $x^2 + y^2 < 1$
B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$
C) P is not the origin

D) $x + y = 1$

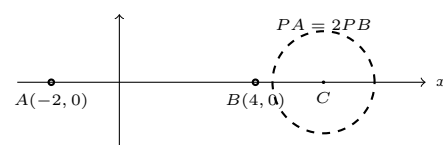
8. The locus of P with $PA = PB$ where $A = (0, 0)$ and $B = (6, 0)$ is the perpendicular bisector $x = 3$. Write down the distance from $(3, 0)$ to $(0, 0)$.
9. A circle has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Write down its centre.
10. Write down the radius of the circle $x^2 + y^2 - 4x + 6y + 9 = 0$.

Section C Substitution & Structure

(≤ 75 s each)

Hard synthesis: square the ratio for Apollonius, test necessary vs sufficient, and spot the hidden betweenness assumption. All Section C are MCQ.

1. The locus of P with $PA = 2PB$ and $A = (-2, 0)$, $B = (4, 0)$ is a circle (Apollonius). What is its centre? (after TMUA 2022 Paper 1 Q7)



- A) $(6, 0)$
 B) $(8, 0)$
 C) $(10, 0)$
 D) $(2, 0)$

2. The locus of points equidistant from the lines $L_1 : x + y = 0$ and $L_2 : x - y = 0$ is: (after TMUA 2018 Paper 2 Q5 / TMUA 2022 Paper 2 Q11)

- A) $y = x$
 B) $y = -x$
 C) $y = |x|$
 D) $x = 0$ or $y = 0$

3. Fil argues: “Let D lie on segment AB and CD bisect $\angle C$. By the angle-bisector theorem $AD/DB = CA/CB$; hence any point P with $PA = PB$ must lie on the bisector of $\angle C$.” Where is the flaw? (after TMUA 2019 Paper 2 Q14)

- A) The proof assumes D lies *between* A and B (betweenness/convexity) without establishing it; D could lie on the extension of AB outside the segment.
 B) Squaring the distance ratio loses the sign, so the locus is a single point not a circle.
 C) The angle-bisector theorem is false for obtuse triangles.
 D) There is no flaw; the conclusion is correct.

4. $PQRS$ is a parallelogram labelled anticlockwise. Consider: I. $PQ = QR$ II. $PR \perp QS$ III. $\angle PQR = 90^\circ$. Which of these conditions is/are individually *sufficient* for $PQRS$ to be a square? (after TMUA 2020 Paper 2 Q7)

- A) I only
 B) II only
 C) III only

- D) I and III only
- E) none of them individually (all insufficient alone)
5. In $\triangle PQR$, $PR = 4$, $QR = p$ and $\angle RPQ = 30^\circ$. For which values of p does this information uniquely determine the length PQ ? (after TMUA 2018 Paper 2 Q14)
- A) $p = 2$
- B) $p = \sqrt{3}$
- C) $1 \leq p < 2$
- D) $p \geq 4$
- E) $p = 2$ or $p \geq 4$
- F) $p = \sqrt{3}$ or $p \geq 4$
- G) $p < 4$
- H) $p \geq 2$
6. The circles $C_1 : (x + 3)^2 + (y - 1)^2 = 25$ and $C_2 : (x - 8)^2 + (y - 1)^2 = 9$ have radii 5 and 3. How many common tangents do they have? (after TMUA 2019 Paper 1 Q6)
- A) 0
- B) 1
- C) 2
- D) 3
- E) 4
7. The Apollonius circle $PA = 3PB$ with $A = (0, 0)$, $B = (6, 0)$ is $(x - \frac{27}{4})^2 + y^2 = (\frac{9}{4})^2$. What is the greatest distance from the origin to a point on this circle? (after TMUA 2022 Paper 1 Q7)
- A) 9
- B) 7.5
- C) 12
- D) 6
8. Fil finds the circle of Apollonius $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$ by writing $(x - 2)^2 + y^2 = 9$ and claiming this is the locus. Where is the error? (after TMUA 2019 Paper 2 Q14)
- A) His equation is a circle centred at B with radius 3, but the true locus from $x^2 + y^2 = 9((x - 2)^2 + y^2)$ is $(x - \frac{9}{4})^2 + y^2 = \frac{9}{16}$ (centre $(\frac{9}{4}, 0)$, radius $\frac{3}{4}$); it incorrectly ignores the distance to A .
- B) Squaring loses the sign information, so the locus is really a single point.
- C) The set $PA = 3PB$ is a straight line, not a circle at all.
- D) There is no error; $(x - 2)^2 + y^2 = 9$ is correct.

Section D Challenge Ramp

(≤ 90 s each)

Hardest 5: stack the week's tools — Apollonius, radical axes, area loci and geometric logic — under TMUA/BMO1 pressure.

1. The locus of P with $PA^2 + PB^2 = 34$, where $A = (-3, 0)$, $B = (3, 0)$, is a circle. What is its radius? (after BMO1 2017 Q4 / TMUA 2022 Paper 1 Q7)
 - A) $2\sqrt{2}$
 - B) 4
 - C) $\sqrt{8}$
 - D) $\sqrt{34}$
2. The graph $|x| + |y| = 3$ is a square (diamond) with vertices $(3, 0)$, $(0, 3)$, $(-3, 0)$, $(0, -3)$. What is the finite area it encloses? (after TMUA 2021 Paper 1 Q9)
 - A) 6
 - B) 12
 - C) 18
 - D) 9
3. The diagram shows a kite $PQRS$ whose diagonals meet at O with $OP = x$, $OQ = y$, $OR = x$, $OS = z$. Which condition is *necessary and sufficient* for $\angle SPQ = 90^\circ$? (after TMUA 2022 Paper 2 Q11)
 - A) $x = y = z$
 - B) $2x = y + z$
 - C) $x^2 = yz$
 - D) $y = z$
 - E) $y^2 = x^2 + z^2$
4. The circles $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - 4)^2 + y^2 = 9$ intersect at P, Q . What is the length PQ ? (after BMO1 2017 Q4 / TMUA Practice Paper 2 Q9)
 - A) 3
 - B) 6
 - C) $\frac{4\sqrt{14}}{3}$
 - D) $\frac{24}{5}$
5. Let the Apollonius circle $PA = 2PB$ with $A = (0, 0)$, $B = (6, 0)$ be $(x - 8)^2 + y^2 = 16$. The triangle PAB varies as P runs over this circle. What is its maximum possible area? (after TMUA 2022 Paper 1 Q7 / BMO1 2017 Q4)

“A locus is a sentence the points all agree on; a geometric proof is only as strong as the condition you actually used.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths